

Supplemental Table 1: Information for data extraction and subsequent summary and appraisal.
Adapted from the CHARMS Checklist and PROBAST risk of bias tool. VL: visceral leishmaniasis, HIV: human immunodeficiency virus

Domain	Key Items
Source of data	Source of data (e.g., cohort, case-control, randomised trial participants, registry data, etc.)
Participants	Participant eligibility and recruitment method (e.g., location, number of centres, setting, inclusion and exclusion criteria)
	Participant description (age, sex, primary VL or relapse case, co-morbidities including HIV co-infection)
	Details of treatments received
	How VL diagnosis is defined (whether consistent for all participants, using serology and/or microscopy, molecular testing, clinical history and physical signs, etc)
	Study dates
Outcome(s) to be predicted	Type of outcome (e.g., single or combined endpoints)
	Definition and method for measurement of outcome (for example, is mortality disease-specific or all-cause, is cure/initial failure/relapse diagnosed based on clinical symptoms and/or diagnostic testing)
	Was the same outcome definition (and method for measurement) used in all patients?
	Time of outcome occurrence or summary of duration of follow-up
	Was the outcome assessed without knowledge of the candidate predictors (i.e., blinded)?
Candidate predictors	Number and type of predictors (e.g., demographics, patient history, physical examination, laboratory parameters, HIV status, disease characteristics, etc)
	Definition and method for measurement of candidate predictors (including whether defined and measured in a similar way for all participants)
	Timing of predictor measurement (e.g., at patient presentation, at diagnosis, at treatment initiation or otherwise)
	Handling of predictors in the modelling (e.g., continuous, linear, non-linear transformations or categorised)
Sample size	Number of participants and number of outcomes/events
	Events per candidate predictor
	Whether the authors describe a sample size calculation

Missing data	Number of participants with any missing value (including predictors and outcomes)
	Number of participants with missing data for each predictor
	Handling of missing data (e.g., complete-case analysis, imputation, or other methods)
Model development	Modelling method (e.g., logistic, survival, or other)
	Modelling assumptions satisfied
	Description of participants that were excluded from the analysis with justification
	Method for selection of predictors for inclusion in multivariable modelling (e.g., all candidate predictors, pre-selection based on unadjusted association with the outcome)
	Method for selection of predictors during multivariable modelling (e.g., full model approach, backward or forward selection) and criteria used (e.g., p-value, Akaike information criterion)
	Shrinkage of predictor weights or regression coefficients (e.g., no shrinkage, uniform shrinkage, penalized estimation)
Model performance	Calibration (calibration plot, calibration slope, Hosmer-Lemeshow test), discrimination (C-statistic, D-statistic, log-rank), and overall performance measures with confidence intervals
	Classification measures (e.g., sensitivity, specificity, predictive values, net reclassification improvement) and whether a priori cut points were used
Model evaluation	Method used for testing model performance: development dataset only (apparent performance, random split of data, resampling methods, e.g., bootstrap or cross-validation, none) or separate external validation
	For external validations; data source and participants to be described as per 'source of data' and 'participants' domains. Definitions and distributions (including missing data) of outcome and candidate predictors.
	In case of poor external validation, whether model was adjusted or updated (e.g., intercept recalibrated, predictor effects adjusted, or new predictors added)
Results	Final and other multivariable models presented, including predictor weights or regression coefficients, intercept, baseline survival, model performance measures (with standard errors or confidence intervals)
	Any alternative presentation of the final prediction models, e.g., sum score, nomogram, score chart, predictions for specific risk subgroups with performance

	Comparison of the definition and distribution of predictors (including missing data) for development and validation datasets
Interpretation and Discussion	Study authors' interpretation of presented models (intended use, clinical utility, etc)
	Study authors' reported strengths and limitations
Miscellaneous	Source of funding / sponsor
	Any declared conflicts of interest
	Methodological guidelines used